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Review article

Multisensory subtypes of aphantasia: Mental imagery as supramodal perception in reverse

Alexei Joel Dawes [a,*], Rebecca Keogh [b], Joel Pearson [c]

[a] RIKEN Center for Brain Science (CBS), Saitama, Japan
[b] School of Psychological Sciences, Macquarie University, Sydney, Australia
[c] School of Psychology, University of New South Wales, Sydney, Australia

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ABSTRACT

Cognitive neuroscience research on mental imagery has largely focused on the visual imagery modality in unimodal task contexts. Recent studies have uncovered striking individual differences in visual imagery capacity, with some individuals reporting a subjective absence of conscious visual imagery ability altogether ("aphantasia"). However, naturalistic mental imagery is often multi-sensory, and preliminary findings suggest that many individuals with aphantasia also report a subjective lack of mental imagery in other sensory domains (such as auditory or olfactory imagery). In this paper, we perform a series of cluster analyses on the multi-sensory imagery questionnaire scores of two large groups of aphantasic subjects, defining latent sub-groups in this sample population. We demonstrate that aphantasia is a heterogenous phenomenon characterised by dominant sub-groups of individuals with visual aphantasia (those who report selective visual imagery absence) and multi-sensory aphantasia (those who report an inability to generate conscious mental imagery in any sensory modality). We replicate our findings in a second large sample and show that more unique aphantasia sub-types also exist, such as individuals with selectively preserved mental imagery in only one sensory modality (e.g. intact auditory imagery). We outline the implications of our findings for network theories of mental imagery, discussing how unique aphantasia aetiologies with distinct self-report patterns might reveal alterations to various levels of the sensory processing hierarchy implicated in mental imagery.

1. Introduction

One interesting phenomenal property of subjective experience is that it is intrinsically multi-sensory. At any one point in time, we interface with a mass of electromagnetic waves, chemical signals, and pressure gradients. Highly specialized receptors in sensory organs (such as our eyes, ears, and skin) measure these afferent stimuli in the world, and this information then travels downstream to dedicated unimodal and hierarchical sensory processing regions (such as the visual or auditory cortices). Eventually, all this unimodal information must be somehow bound together temporally and contextually to allow for coherent, actionable, and unified knowledge of the current environment. However, in addition to multi-sensory perceptual experience, coherent multi-sensory mental simulations in the total absence of afferent perceptual input can also occur, such as during mental imagery. When daydreaming about a sunset at the beach, we might "see" the yellow glow of the setting sun, "hear" the sound of waves crashing on the shore, and "smell" the salt in the sea spray, all purely within the mind. Through mental imagery, these elements are subjectively experienced as a coherent and unified multisensory scene, offering a flexible and powerful tool to aid memory and decision-making.

Although endogenous, these sensory representations are far from illusory or merely propositional, and mental imagery is known to elicit content-specific activity throughout the sensory processing hierarchy. Visual imagery can be represented using a depictive format, engaging area V1 in a retinotopically and orientation-specific manner similar to visual perception (Pearson et al., 2008; Pearson and Kosslyn, 2015; Slotnick et al., 2005). Visual imagery can also be successfully cross-decoded in early visual cortex by models trained on equivalent perceptual stimuli in both single-feature (Albers et al., 2013; Kamitani and Tong, 2006; Koenig-Robert and Pearson, 2019; Stokes et al., 2009) and multi-feature (Huang et al., 2023; Naselaris et al., 2015) encoding contexts. Later in the visual processing hierarchy, striking neural overlap between imagery and perception is seen in category-specific

* Corresponding author.
E-mail address: alexeijoeldawes@gmail.com (A.J. Dawes).

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occipital regions such as the fusiform face area (FFA) and parahippocampal place area (PPA; Ishai et al., 2002; O'Craven and Kanwisher, 2000), as well as in scene-selective subregions of precuneus (Johnson and Johnson, 2014).

Similar results attesting to stimulus-specific activation of unimodal primary sensory regions during mental imagery have also been demonstrated independently in the auditory (Hubbard, 2010; Proix et al., 2022; Zatorre and Halpern, 2005), motor (Ehrsson et al., 2003; Kasai et al., 1997), somatosensory/tactile (Nierhaus et al., 2023; Yoo et al., 2003), and even gustatory/taste (Avery et al., 2023) domains. This implies that mental imagery may work in a similar way across sensory modalities, with higher association or executive control regions in the brain exerting top-down control over primary sensory regions to shape stimulus-specific imagery representations according to task goals. However, because many existing studies of mental imagery often justifiably rely on single-feature stimuli in unimodal sensory domains, it is unclear whether multi-sensory mental imagery should work in the same way across modalities. Evidence that visual imagery and visual perception share a common architecture in primary visual regions, for example, does not intrinsically suggest that naturalistic mental imagery is unimodal or constrained to the visual domain. Multi-sensory imagery may, in fact, be the norm rather than the exception for human cognition. How, then, do our brains flexibly recombine sensory representations from independent and distributed unimodal regions to allow for multi-sensory integration during mental imagery? When we daydream about that beach sunset, how are the imagined sights, sounds, and smells stitched together to form a coherent multi-sensory scene?

One useful empirical approach to the question of how we come to subjectively experience multisensory mental imagery is to examine subjective experience itself. Perhaps the most profound example of the extent to which subjective imagery vividness can vary is a rare phenomenon known as aphantasia (Zeman et al., 2015; Zeman et al., 2010). Individuals with aphantasia subjectively report a total congenital lack of conscious visual imagery in the absence of any neuropsychiatric history (Dawes et al., 2020; Zeman et al., 2020). Perplexingly, these individuals often perform typically on visual working memory tasks (Keogh, Wicken et al., 2021; Pounder et al., 2022; Zhao et al., 2022), but do show stark differences in naturalistic imagery paradigms such as autobiographical memory tasks (Dawes et al., 2022; Milton et al., 2021). Although metacognitive mechanisms may contribute in part to the subjective absence of visual imagery in aphantasia, individuals from this sub-population do perform worse than control subjects on standardised behavioural measures of visual imagery strength (Keogh and Pearson, 2018) and do not exhibit physiological markers of visual imagery processes such as a pupillometric signature of imagined luminance (Kay et al., 2022) and a skin conductance response to fear imagery (Wicken et al., 2021) that are otherwise apparent in individuals who report having visual imagery.

To date, aphantasia has predominantly been categorised on the basis of self-reported visual imagery vividness scores alone (or a combination of self-report and behavioural visual imagery measures; Wicken et al., 2021), and has most commonly been explored as a visual phenomenon. This is likely due to the nature of the modern operationalisation of aphantasia, a term coined on the basis of the experience of a patient who reported losing their ability to visualise after surgery (Zeman et al., 2010). However, since the coining of this term, many individuals with aphantasia have reported an inability to imagine in other sensory modalities as well. We have previously suggested that aphantasia may underpinned by unique aetiological profiles and heterogenous sub-groups (Dawes et al., 2020), showing that around 26% of aphantasic individuals who completed our online study reported a total absence of mental imagery in all sensory modalities (not just visual) on the Questionnaire for Mental Imagery (Sheehan, 1967). These results are supported by more recent findings that individuals with aphantasia also disproportionately report auditory imagery deficits (Hinwar and Lambert, 2021) and multi-sensory imagery deficits (Dance et al., 2021), as well as by currently unpublished work which appears to make a similar observation of distinct sub-populations in aphantasia who report either multi-sensory imagery absence or visual imagery absence alone (Takahashi et al., 2023).

Understanding population-level patterns of multi-sensory imagery reports is important because it may help to both reveal distinct developmental aetiologies for aphantasia and shed light on mechanisms for multi-sensory integration during mental imagery. If a majority of individuals with aphantasia selectively report visual imagery deficits, this would provide preliminary evidence that aphantasia may indeed be primarily a visual phenomenon emerging from anatomical correlates in occipital brain regions (Bergmann et al., 2016; Cui et al., 2007; Keogh et al., 2020). On the other hand, if a majority of individuals with aphantasia in fact report multi-sensory imagery deficits across the board, this may imply that aphantasia is instead a more generalized phenomenon characterized by deficits in top-down control mechanisms, multi-sensory integration processes, or higher-order metacognitive mechanisms. If such a generalized phenomenon (multi-sensory aphantasia or multi-sensory "mind blindness") were relatively common in mental imagery, it would also stand in stark contrast to the lack of such a phenomenon in perception. This is because no known condition of full multi-sensory perceptual "blindness" or multi-sensory processing impairment has been documented – although unilateral spatial neglect is a close analogy, it is better thought of as a disorder of visual attention (and cross-modal sensory integration often remains intact for individuals with this condition; Vallar and Bolognini, 2014). Hence, if such a condition did occur by analogy in mental imagery, this would mark an important point of difference between imagery and perception and reinforce that they are underpinned by fundamentally different architectures. Specifically, multi-sensory aphantasia (if it exists) might indicate alterations to a supramodal process leading to the attenuation or abolishment of all imagined senses in a manner that would be exceedingly rare if it occurred during regular perception. A robust estimate of the prevalence of multi-sensory aphantasia can therefore provide us with useful information about the system differences between mental imagery and sensory perception.

In the current study, we use a data-driven comparative clustering approach to systematically categorise patterns of self-reported mental imagery deficits across sensory modalities in large samples of individuals reporting aphantasia. We demonstrate that aphantasia is indeed a condition characterised by heterogeneous sub-groups: those with visual aphantasia, those with multi-sensory aphantasia, and those with more nuanced patterns of multi-sensory imagery deficits.

2. Methods and materials

2.1. Participants

Two independent large samples of aphantasic subjects were collated during a series of online studies at the University of New South Wales (UNSW; Sydney, Australia). All subjects self-reported aphantasia and had either approached the lab directly for the first time or were recruited from online community platforms. All collated studies (including recruitment via relevant platforms) were approved by the UNSW Human Research Ethics Advisory Panel (HREAP-C) in compliance with ethical human research guidelines set by the National Health and Medical Research Council. Given the exploratory nature of our analyses, we sought to maximise our sample size based on the number of subjects across our internal databases who had completed both a visual imagery questionnaire (Vividness of Visual Imagery Questionnaire; Marks, 1973) and a multi-sensory imagery questionnaire (Questionnaire upon Mental Imagery; Sheehan, 1967). Each scale was completed as part of a battery of questionnaires administered in random order at the beginning of each relevant online study. In addition to the questionnaires above, subjects in Sample 1 took part in one of four studies investigating the relationship between visual imagery and memory or decision-making (not reported here). Subjects in Sample 2 took part in a single online study

investigating individual differences in visual imagery and self-reported cognition. There was no overlap between our sample populations, and no included subject had completed more than one of the relevant studies. Our final sample sizes were as follows: Sample 1 ($N = 964$; mean age = 39.24 ± 15.16 *SD*; 542 female, 10 gender non-binary, 32 prefer not to say) and Sample 2 ($N = 1148$; mean age = 40.00 ± 16.82 *SD*; 775 female, 21 gender non-binary, 2 prefer not to say).

2.2. Questionnaires

The Vividness of Visual Imagery Questionnaire (VVIQ; Marks, 1973) is the most common assessment tool used to categorise aphantasia in the mental imagery literature. Subjects generate visual imagery according to item prompts and self-rate their visual imagery vividness on 16 items using a Likert scale from 1 ("No image at all, you only 'know' that you are thinking of the object") to 5 ("Perfectly clear and as vivid as normal vision"). We computed a sum score across all 16 items, with a score of 16 representing very weak self-reported imagery, and a score of 80 representing very vivid self-reported imagery. The mean VVIQ sum scores for each sample were as follows: Sample 1 (mean VVIQ sum score = 17.71 ± 3.49 *SD*) and Sample 2 (mean VVIQ sum score = 18.25 ± 3.84 *SD*).

All subjects included in the analysis also completed a multi-sensory imagery questionnaire, the Questionnaire upon Mental Imagery (Sheehan, 1967). Subjects imagine stimuli in seven sensory modalities (visual, auditory, tactile, kinesthetic, olfactory, gustatory, and somatic) and rate the clarity and vividness of their imagery on a 7-point Likert-type scale ranging from 1 ("I think of it, but do not have an image before me") to 7 ("Very vivid and as clear as reality"), using this same rating scale on every item. There are five items per sensory modality, and 35 QMI items in total. Note that although the seventh QMI sensory modality is often called "emotion imagery" in the literature, this is not an accurate description of this modality's scale items which more closely reflect embodied or somatic states such as hunger (e.g. "how clear and vivid can you imagine the feeling of satisfaction after a tasty meal?") or bodily discomfort (e.g. "how clear and vivid can you imagine the feeling of sore throat?"). To reflect this, we have termed the modality "somatic imagery" here. A sum score was computed separately for each sensory modality for each subject, such that a subject's modality-specific scores could range from 5 (representing a lack of subjective mental imagery in that modality) to 35 (representing very strong imagery in that modality). The mean QMI modality sum scores for each sample were as follows: visual imagery (Sample 1 mean sum score = 5.94 ± 2.25 *SD;* Sample 2 mean sum score = 6.1 ± 2.99 *SD*); auditory imagery (Sample 1 mean sum score = 13.28 ± 9.84 *SD;* Sample 2 mean sum score = 12.79 ± 9.24 *SD*); tactile imagery (Sample 1 mean sum score = 11.3 ± 8.63 *SD;* Sample 2 mean sum score = 11.06 ± 8.35 *SD*); kinesthetic imagery (Sample 1 mean sum score = 12.08 ± 8.99 *SD;* Sample 2 mean sum score = 11.96 ± 8.68 *SD*); taste imagery (Sample 1 mean sum score = 10.43 ± 8.26 *SD;* Sample 2 mean sum score = 10.75 ± 8.49 *SD*); olfactory imagery (Sample 1 mean sum score = 9.77 ± 7.59 *SD;* Sample 2 mean sum score = 10.16 ± 7.93 *SD*); somatic imagery (Sample 1 mean sum score = 14.38 ± 9.96 *SD;* Sample 2 mean sum score = 14.27 ± 9.58 *SD*). There were no significant differences between Aphantasia Sample 1 and Aphantasia Sample 2 in their mean sum scores on any QMI modality (all $t < 1.32$, all $p > .05$).

2.3. Clustering analyses

We conducted a series of exploratory clustering analyses to classify latent sub-groups in aphantasia based on the multi-sensory imagery reports (sum scores in each modality of the QMI) of our large subject samples. Without ground truth on the number of underlying sub-groups that might exist, we took a conservative approach by comparing agreement across three unsupervised cluster algorithms from different classes: a centroid model (the k-means algorithm), a probabilistic distribution model (Gaussian mixture modelling; GMM), and a density model (Density-based spatial clustering of applications with noise; DBSCAN). All models were conducted in Python 3.8.12 using the Scikit-learn module (Pedregosa et al., 2011). All three models were fitted separately for Aphantasia Sample 1 ($N = 964$) and Aphantasia Sample 2 ($N = 1148$) using identical parameters to compare the replicability of the cluster estimates across independent sample populations.

2.3.1. Clustering algorithm parameters

For DBSCAN, we set the minimum samples at 10 (equivalent to the number of subjects in a neighborhood for a point in that neighbourhood to be considered a core point) and epsilon at 0.5 (equivalent to the maximum Euclidean distance between two subjects for them to be considered as being in overlapping neighbourhoods). Other DBSCAN parameters were set at default. For both k-means and GMM, 20 independent algorithms were fitted to each dataset (corresponding to the number of fixed clusters to be formed for each algorithm, ranging from 2 to 20). For each k-means algorithm, a maximum of 300 iterations was performed; for each GMM algorithm, a maximum of 100 iterations was performed. For both k-means and GMM, 10 initializations were performed for each iteration (meaning that the algorithm was run 10 times per iteration using different centroids, with the best initialization chosen as output for that iteration). For each algorithm, we computed a Silhouette co-efficient based on Euclidean distance, with higher Silhouette scores indicating an algorithm with better defined clusters. We chose the algorithm with the highest Silhouette score as our optimal solution for that algorithm class (k-means or GMM). Plots of Silhouette scores relative to the defined number of clusters for each algorithm class (k-means and GMM) and sample population (Aphantasia Sample 1 and Aphantasia Sample 2) can be found in Supplementary Materials (Figs. S1–S4).

2.3.2. Clustering algorithm agreement

For each sample population, we categorised subjects into four primary groups based on label agreement across our three clustering algorithm classes (k-means, GMM, and DBSCAN). A multi-sensory aphantasia group featured subjects with floor scores on all QMI sensory modalities where all three clustering algorithm solutions agreed on the same group label. A "weak" multi-sensory aphantasia group reflected subjects with low to floor scores on all QMI sensory modalities where only two out of three clustering algorithm classes agreed on the group label for a given subject. A visual aphantasia group featured subjects with floor scores on the visual QMI modality but moderate to high scores on other QMI sensory modality domains, with full agreement across all three clustering algorithms. Lastly, a "weak" visual aphantasia group included subjects resembling those in the visual aphantasia group, but where only two out of three clustering algorithms agreed on the group label for a given subject.

2.3.3. Data visualization

To visualize major aphantasia sub-groups identified by our clustering analyses, we first performed dimensionality reduction on the QMI data (sum scores in seven sensory modalities). T-distributed Stochastic Neighbour Embedding (T-SNE; van der Maaten and Hinton, 2008) was performed on the 7 QMI modalities using the Scikit-learn manifold module (Pedregosa et al., 2011) in Python 3.8.12. Unlike linear dimension reduction methods such as Principal Components Analysis (PCA), T-SNE is a non-linear dimension reduction algorithm that seeks to preserve a balance of local and global structure in high-dimensional data. T-SNE was performed for data visualization only, to allow us to portray the group labels output by our clustering analyses in a compressed two-dimensional space. All clustering analyses, however, were performed on untransformed data. We also used violin plots (with median and interquartile range) to depict the distribution of untransformed sum scores in each QMI sensory imagery modality across aphantasia sub-groups and sample populations. The overlap in clustering algorithm agreement on group labels was depicted using simple stacked bar plots.

3. Results

In the current study, we conducted a data-driven classification of the sub-types of aphantasia, a phenomenon marked by the subjective absence of visual mental imagery. Cluster analyses from three algorithm classes (k-means, GMM, and DBSCAN) were performed on the untransformed multi-sensory imagery questionnaire (QMI) scores from two large independent samples of aphantasic subjects. Twenty algorithms were fitted for the k-means and GMM classes (corresponding to tested cluster solutions from two clusters through to twenty clusters), with the best cluster solution in each case being chosen based on the highest output Silhouette score.

3.1. Visual aphantasia and multi-sensory aphantasia

Fig. 1a shows the T-SNE embedding (dimension reduction) of the data from our Aphantasia Sample 1 ($N = 964$) with corresponding cluster labels. In this sample, all three cluster algorithm classes (k-means, GMM, and DBSCAN) produced a two-cluster solution for the QMI data. The maximum Silhouette score corresponding to this two-cluster solution was 0.529 for k-means and 0.295 for GMM. The two-cluster solution produced by each algorithm class definitively corresponded to a multi-sensory aphantasia group (Fig. 1a; yellow cluster) with floor scores on all imagery modalities of the QMI (Fig. 1b, top-left panel; yellow plots), and a visual aphantasia group (Fig. 1a; light green cluster) who scored at floor on the visual imagery sensory modality of the QMI (Fig. 1b, bottom-right panel; light green plots), but reported retained imagery in all other sensory modalities. There was partial agreement by each clustering algorithm on the group label assignment for these two clusters (the corresponding estimates from each cluster algorithm can be found in Table S1 (Supplementary Materials). For Aphantasia Sample 1 ($N = 964$), all three cluster algorithms agreed that 233 subjects (24.2%) belonged to the multi-sensory aphantasia group, and that 308 subjects (31.95%) belonged to the visual aphantasia group.

We categorised the remaining subjects based on the extent of agreement across the three cluster algorithm classes. For the Aphantasia Sample 1, there were 192 subjects (19.92%) assigned by k-means and GMM (but not DBSCAN) to a multi-sensory aphantasia cluster (which we termed "weak" multi-sensory aphantasia; Fig. 1a and c, red data). Similarly, there were 231 subjects (23.96%) assigned by GMM and DBSCAN (but not by k-means) to a visual aphantasia cluster (which we termed "weak" visual aphantasia; Fig. 1a and c, blue data). Fig. 1b demonstrates that although the QMI scores of the "weak" visual aphantasia group (blue violin plots, bottom-left panel) still reflect weak subjective multi-sensory imagery reports, their median QMI scores in each non-visual sensory imagery modality (marked by thick lines in each violin plot) are qualitatively greater than those for the "weak" multi-sensory aphantasia group (Fig. 1b, red violin plots, top-right panel) whose group median in each modality is usually at floor. Fig. 1b (yellow violin plots, top-left panel) also illustrates that the subjects assigned by all three clustering algorithm classes to the primary multi-sensory aphantasia cluster in Aphantasia Sample 1 are essentially all subjects who scored at floor (5 out of a possible 35) on each modality of the QMI, with no variance in the scores for this group. These subjects invariably report a total absence of multi-sensory imagery. The proportion estimate for the combined multi-sensory aphantasia cluster (Fig. 1a - c, yellow data) and the "weak" multi-sensory aphantasia cluster (Fig. 1a - c, red data) in Aphantasia Sample 1 ($N = 964$) is 425 subjects (44.09%). The proportion estimate for the combined visual aphantasia (Fig. 1a - c, light green data) and the "weak" visual aphantasia group (Fig. 1a - c, blue data) in Aphantasia Sample 1 ($N = 964$) is 539 subjects (55.91%).

3.2. Clustering replicability across independent samples

Next, to ensure that our aphantasia categories and sample proportion estimates were reliable, we assessed a new and independent sample of subjects reporting visual aphantasia (Aphantasia Sample 2; $N = 1148$). Fig. 1d and e demonstrate the replicability of our group estimates, showing that the results of our independent clustering analyses were very similar in our second Aphantasia Sample ($N = 1148$). Here too, all three cluster algorithm classes (k-means, GMM, and DBSCAN) produced a two-cluster solution for the QMI data corresponding to sub-groups of individuals with multi-sensory aphantasia (yellow data) and visual aphantasia (light green data). The maximum Silhouette score corresponding to the two-cluster solution in Aphantasia Sample 2 was 0.536 for k-means and 0.231 for GMM. Based on complete agreement across the three clustering algorithm classes, the proportion of subjects assigned to each of the two group labels was very similar in Aphantasia Sample 2 (276 subjects comprising 24.02% of the sample in the multi-sensory aphantasia group, and 348 subjects comprising 30.31% of the sample in the visual aphantasia group) relative to Aphantasia Sample 1. However, Fig. 1f illustrates that there were slight differences in the overall patterns of agreement across clustering algorithms in Aphantasia Sample 2. Here, only 111 subjects (9.67%) fell under our "weak" multi-sensory aphantasia group (where both k-means and GMM, but not DBSCAN, agreed on the multi-sensory aphantasia cluster label) compared to 19.92% in Aphantasia Sample 1); and 413 subjects (35.96%) fell under the "weak" visual aphantasia group (where both GMM and DBSCAN, but not k-means, agreed on the visual aphantasia cluster label) compared to 23.96% in Aphantasia Sample 1.

Comparing Fig. 1c and f, the overall proportion estimates for the primary multi-sensory aphantasia and visual aphantasia clusters (based on complete agreement across algorithms) are broadly similar across Aphantasia Samples 1 and 2 because they effectively correspond to the DBSCAN multi-sensory aphantasia estimate, and the k-means visual aphantasia estimate, respectively, in each independent sample. The comparatively lower aggregated proportion estimate formed for the "weak" multi-sensory aphantasia group in Aphantasia Sample 2 (Fig. 1f, Agree column, red stacked bar) simply reflects a more conservative multi-sensory aphantasia cluster label assignment by the GMM algorithm in Aphantasia Sample 2 relative to Aphantasia Sample 1. This in turn affects the combined proportion estimates for multi-sensory aphantasia and visual aphantasia in Aphantasia Sample 2. Here, the proportion estimate for the combined multi-sensory aphantasia cluster (Fig. 1d - f, yellow data) and the "weak" multi-sensory aphantasia cluster (Fig. 1d - f, red data) in Aphantasia Sample 2 ($N = 1148$) is 387 subjects (33.71%, compared to 44.09% in Aphantasia Sample 1). The proportion estimate for the combined visual aphantasia (Fig. 1d - f, light green data) and the "weak" visual aphantasia group (Fig. 1d - f, blue data) in Aphantasia Sample 2 ($N = 1148$) is 761 subjects (66.29%, compared to 55.91% in Aphantasia Sample 1).

3.3. Heterogeneous sub-groups in aphantasia

Our primary clustering analyses revealed that in both independent aphantasia samples, there were large sub-classes of individuals reporting pure visual aphantasia (at least 30% of subjects) and multi-sensory aphantasia (absence of imagery across all modalities; at least 24% of subjects). Approximately half of the remaining individuals in each sample group (the "weak" visual aphantasia and "weak" multi-sensory aphantasia groups, collectively) produced patterns of scores on the QMI for which the three clustering algorithm classes (k-means, GMM, and DBSCAN) did not reach consensus label assignment. If these weak agreement groups feature highly heterogeneous patterns of variance in QMI scores, re-subjecting those groups alone to additional clustering is unlikely to yield interpretable results with consensus agreement across clustering algorithms. However, we were still interested in whether the aphantasia population as a whole (with the exception of the already defined primary groups of visual aphantasia and multi-sensory aphantasia) might feature small but interesting sub-groups of individuals with clear patterns of selective aphantasia (such as preserved auditory imagery, for example, or selectively impaired visual and auditory

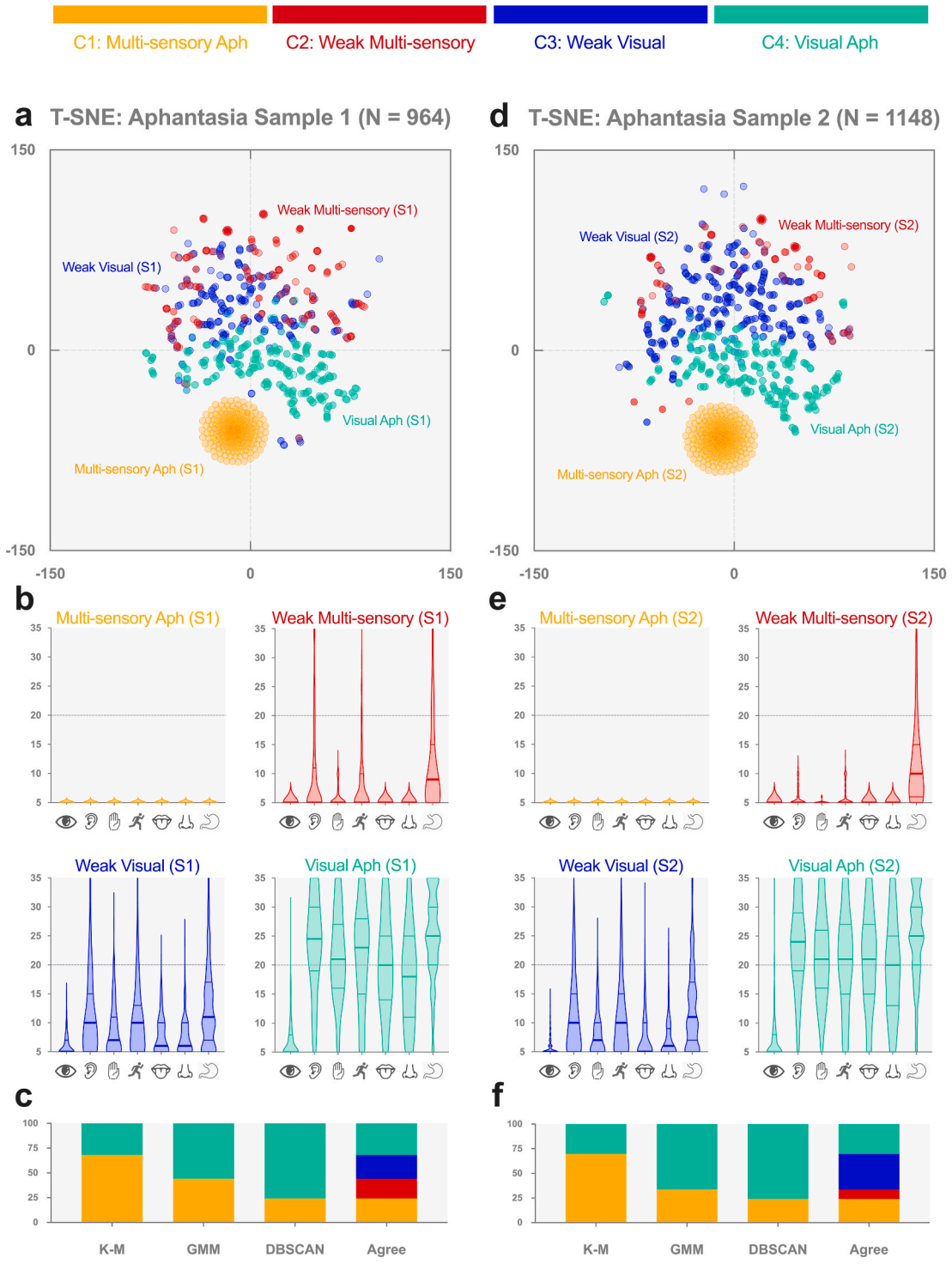


Fig. 1. Results of comparative clustering analyses on the QMI scores of independent aphantasia samples. Panels a and d show cluster labels for Aphantasia Sample 1 ($N = 964$) and Aphantasia Sample 2 ($N = 1148$) respectively, by colour (yellow = multi-sensory aphantasia; red = weak agreement multi-sensory aphantasia; light green = visual aphantasia; blue = weak agreement visual aphantasia). Data points represent two-dimensional embedding using T-distributed Stochastic Neighbour Embedding (T-SNE) for data visualization only (all analyses were conducted on untransformed QMI data). Panels b and e show kernel-density smoothed violin plots for each aphantasia sub-group for Aphantasia Sample 1 and Aphantasia Sample 2, respectively, with the QMI modality-specific sum score on the vertical axis (ranging from 5 to 35) and the QMI sensory modality on the horizontal axis (from left to right: visual, auditory, tactile, kinesthetic, gustatory, olfactory, and somatic imagery). Panels c and f show the extent to which separate clustering algorithms agreed on the labels assigned to each group. Here, the horizontal axis shows the respective clustering algorithm's estimate of the percentage (0–100%) of the sample population belonging to each major aphantasia sub-group (light green = visual aphantasia; yellow = multi-sensory aphantasia). For example, in Aphantasia Sample 1 (panel c), the optimum Gaussian Mixture Modelling (GMM) cluster solution assigned 56% of the sample population ($N = 1148$) to the visual aphantasia group and 44% to the multi-sensory aphantasia group. The Agree stacked bar column at far right depicts the extent of agreement across clustering algorithms; for example, in Aphantasia Sample 2 (panel **f**), both the GMM and DBSCAN algorithms (but not k-means) assigned 36% of subjects to the visual aphantasia group; these are the subjects categorised as "weak visual aphantasia" (blue points) in panel d.

imagery).

A full analysis of all possible permutations of QMI modality patterns is not feasible; assuming 7 possible QMI sum scores in each of the 7 QMI modalities, the number of possible sum score combinations across the scale for a given subject is equal to 1716 (derived using the "stars and bars" combinatorics formula). Nevertheless, we took a more straightforward approach by combining both Aphantasia Sample 1 and Aphantasia Sample 2 into a full collated dataset ($N = 2112$) and asking two basic questions: firstly, how many aphantasic individuals have *only one* other sensory imagery modality affected (in addition to visual imagery)? Second, how many aphantasic individuals have only one selectively *preserved* sensory imagery modality, with all other sensory modalities (including visual imagery) affected? We first set a median cut-off at 20 on each QMI modality (reflecting median scores of 4 out of 7 on single items in a given sensory modality), and then completed a manual search through the 12 possible combinations of QMI score patterns matching our two questions (for example, "auditory-preserved" aphantasia would be revealed in a sub-group of individuals with QMI auditory modality scores above 20, and scores below 20 in all other QMI sensory modalities; and so on).

Our manual combination search revealed that many heterogeneous patterns of multi-sensory imagery reports likely exist in the aphantasia sample population, although none reflect high population variance to the extent of our primary multi-sensory aphantasia (~24%) and visual aphantasia (~30%) groups. A table of our manual search results is provided in Table S2 (Supplementary Materials); only those sub-groups representing greater than 1% of the combined aphantasia sample population are reported here. Fig. 2a shows the two-dimensional T-SNE embedding for all aphantasic subjects in the current study ($N = 2112$). The sub-group reflecting the highest proportion of the combined sample population (5.16% comprising 109 subjects) included individuals reporting intact "somatic" mental imagery (such as imagining the feeling of hunger, or the pain of a sore throat), but weak sensory imagery in all other QMI modalities (Fig. 2a and b, pink data). Other sub-groups included individuals with selectively preserved auditory imagery but impaired sensory imagery in all other modalities (3.74% comprising 74 subjects; Fig. 2b, upper-right panel, dark green data); individuals with selectively preserved kinesthetic imagery but impaired imagery in other modalities (1.75% comprising 37 subjects; Fig. 2b, bottom-left panel, orange data); and individuals with strong mental imagery in all sensory modalities except for the visual and kinesthetic modalities (1.47% comprising 31 subjects; Fig. 2b, bottom-right panel, purple data).

4. Discussion

Here, we examined the multi-sensory imagery reports of a rare subpopulation of individuals who have no conscious experience of visual imagery (individuals with aphantasia). Using a comparative clustering approach (featuring three cluster algorithm classes), we identified two primary sub-groups in aphantasia: a *visual aphantasia* group characterised by reported impairments in visual imagery but intact imagery in all other sensory imagery modalities, and a *multi-sensory aphantasia* group characterised by reported impairment in all sensory imagery modalities. This expands on previous work identifying that many individuals with aphantasia (primarily categorised as a visual phenomenon) report deficits in other sensory mental imagery modalities as well (Dawes et al., 2020; Dance et al., 2021; Hinwar & Lambert, 2021; Takahashi et al., 2023). Here, however, we demonstrate the extent of multi-sensory aphantasia more unequivocally by taking a conservative, data-driven approach and replicating our findings across two large, independent samples. From our current findings, we can now more confidently determine that at least 24% of individuals with aphantasia report no multi-sensory imagery capacity at all in any modality, a population estimate very close to our original approximation (26.22% in Dawes et al., 2020). This estimate has important ramifications for imagery research, where visual imagery is typically studied in a unimodal task context and where aphantasia has traditionally been operationalised (and categorised) solely as a condition of visual imagery absence. Future researchers interested in individual differences in mental imagery ability should consider carefully administering multi-sensory imagery questionnaires such as the QMI (Sheehan, 1967) or other multi-sensory imagery scales (Andrade et al., 2014; Dance et al., 2021).

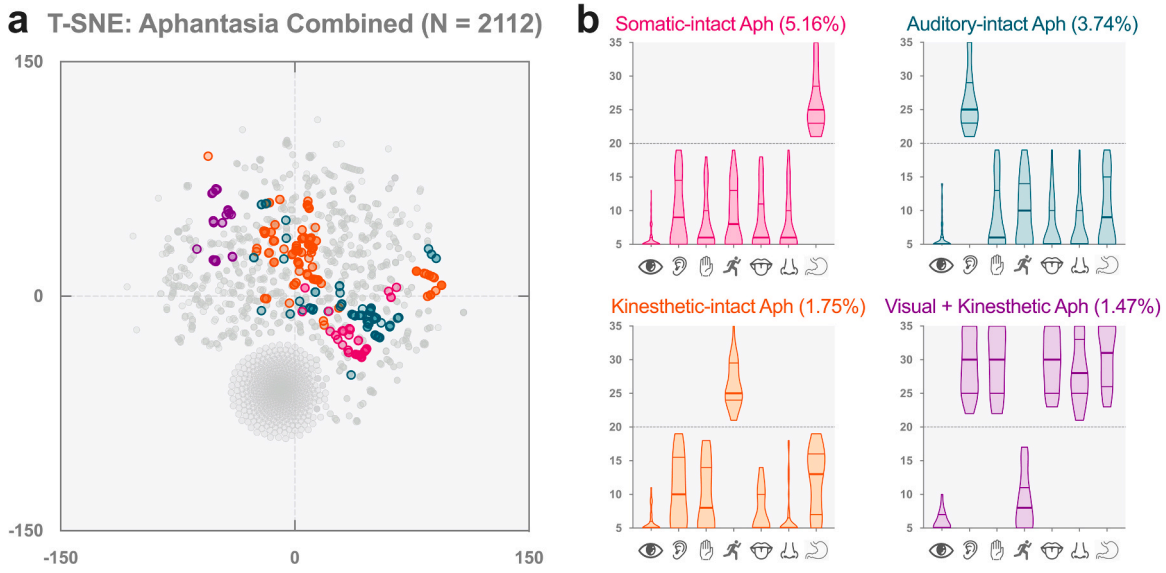


Fig. 2. Results of manual search of some possible combinations of QMI modality score patterns in the collated aphantasia samples (Aphantasia Sample 1 and Aphantasia Sample 2; combined $N = 2112$) using an "above vs. below" median cut-off rule of 20 out of 35 sum score on each QMI modality. Panel **a** shows T-SNE two-dimensional embedding of the individual subject data points ($N = 2112$). Aside from dominant majority sub-groups in aphantasia (those with multi-sensory aphantasia and visual aphantasia, respectively), heterogeneous patterns of multi-sensory imagery reports also exist. Panel **b** shows four such sub-groups (only those representing greater than a 1% proportion of the combined aphantasia sample). These sub-groups include individuals with selectively preserved "somatic" mental imagery (pink data, upper-left panel), those with selectively preserved auditory imagery (dark green data, upper-right panel), those with selectively preserved kinesthetic imagery (orange data, bottom-left panel), and those with preserved mental imagery in all sensory modalities except for the visual and kinesthetic modalities (purple data, bottom-right panel).

We also observed considerable heterogeneity amongst the multi-sensory imagery questionnaire scores of individuals with visual (rather than multi-sensory) aphantasia. Specifically, it is not necessarily the case that a remaining majority (~76%) of aphantasic individuals report a common imagery experience defined by impaired visual imagery and preserved imagery in all other sensory modalities (as might be inferred from previous studies in the field). In our current analyses, the maximum consensus agreement on visual aphantasia group label assignment amongst our three clustering algorithm classes was 30% in Aphantasia Sample 1, and 31% in Aphantasia Sample 2. This indicates that we can only say with full confidence that ~30% of individuals with aphantasia will share a pattern of selectively impaired visual imagery and preserved imagery in all other sensory modalities. To a certain extent, this estimate simply reflects the different mathematical approaches taken toward unsupervised clustering by centroid models (k-means), probability-based distribution models (GMM), and density-based models (DBSCAN). We should not expect these algorithm classes to have given equivalent results, and relying on the converging agreement across all three algorithms is arguably an artificial approach likely to yield an overly conservative estimate of the population prevalence of visual aphantasia. However, it is also clear from Fig. 1b and e that the subjective mental imagery experiences of the "weak" visual aphantasia group (blue violin plots) are categorically different from those of the primary visual aphantasia group (light green violin plots). For example, unlike in the primary visual aphantasia cluster, the group medians for the "weak" visual aphantasia group fall well below the median modality score (20 out of 35), reflecting comparatively weaker multi-sensory imagery in non-visual modalities. In scale terms, and excluding the visual imagery modality, this is the difference between reporting multi-sensory imagery that is "still recognizable" or "generally clear and vivid" (in the primary visual aphantasia cluster) and reporting multi-sensory imagery that is "very vague and hardly recognizable" (in the "weak" visual aphantasia group), according to the QMI rating scale terminology.

Likewise, ignoring agreement across cluster algorithm classes would have meant lumping together the primary multi-sensory aphantasia cluster and the "weak" multi-sensory aphantasia cluster, the latter of which contains individuals reporting intact imagery in at least some non-visual sensory modalities (including somatic imagery in both sample populations). The point is driven home by our second exploratory combination search (Fig. 2), in which we observed at least 12% of the combined aphantasia sample (251 subjects out of 2112) who show unique patterns of multi-sensory imagery reports characterised by selective impairment of only one other non-visual imagery modality (in addition to absent visual imagery) or selective preservation of only one sensory imagery modality. To subsume these unique sub-groups under a single "visual aphantasia" label would be to infer that an aphantasic individual who reports only retaining auditory mental imagery shares the same subjective experience as an aphantasic individual reporting no visual imagery but moderate to strong imagery in other sensory modalities. These cases highlight the advantage of taking a conservative consensus agreement approach when classifying sub-groups in aphantasia; we can state more confidently our estimates of population prevalence whilst attempting to prevent dilution of individual experiences.

There are several potential neural mechanisms thought to underlie mental imagery that might help to place our findings into perspective. One dominant model of mental imagery in the visual domain posits that it acts like a kind of "vision in reverse" (Pearson, 2019). The model says that whilst visual perception involves a feedforward sweep of information from posterior to anterior brain regions (broadly speaking), information flow during visual imagery is instead reversed such that top-down signals originating in fronto-parietal regions trigger a downstream cascade of information that eventually shapes stimulus-specific representations in primary sensory areas (Koenig-Robert and Pearson, 2021). There is general support in cognitive neuroscience for this simple model in the visual domain (Dentico et al., 2014; Dijkstra, Zeidman et al., 2017; Dijkstra et al., 2020). However, it is unclear how this model should apply in the case of multi-sensory imagery, where top-down signals must shape imagery representations in multiple sensory regions (either in concert or in unison).

Evidence of cross-modal imagery interactions may provide some clues. For example, it has been shown that category-specific auditory information induced by imagined sounds transfers to the coding space of early visual cortex in blindfolded subjects, where this abstracted information can be read out using multivariate pattern analysis techniques (Vetter et al., 2014). Imagery alone can elicit classic cross-modal illusions in sensory perception (Berger and Ehrsson, 2013), including the McGurk effect (where auditory imagery biases perception of speech), the cross-bounce illusion (where auditory imagery biases visual perception) and the ventriloquist illusion (where visual imagery biases auditory perception). Using whole-brain searchlight analyses, it is also possible to demonstrate that searchlights trained to distinguish between auditory imagery and tactile imagery of the same cross-modal stimuli (tactile figurines paired by association with auditory pitch) show success not only in primary somatosensory and auditory cortices but also in primary motor and visual cortices as well (de Borst and de Gelder, 2017). Together, these findings imply that mental imagery can indeed be inherently cross-modal, with a common top-down signal eliciting category-specific modulation of multiple primary sensory regions supporting the same imagined multi-sensory stimulus.

This latter evidence would help to explain multi-sensory aphantasia to a certain degree. If mental imagery activates multiple primary sensory regions by default (irrespective of the imagery modality), then perhaps individuals with multi-sensory aphantasia are making veridical reports of a lack of multi-sensory imagery occurring due to weak top-down signals from fronto-parietal regions. The imagery activity in various sensory regions would be weak, not because of multiple structural deficits occurring by chance in all primary sensory cortices, but because the top-down imagery signal would be insufficient to elicit precise sensory representations downstream. This would help to explain why multi-sensory "blindness" is seemingly common during attempted mental imagery in aphantasia, but so rare during perception in the wider population. There are of course cases of multi-sensory impairment in perception isolated to several sensory modalities (as in the case of deaf-blindness; Calvert et al., 2004) or in other instances to an entire unilateral hemifield (as in the case of spatial neglect; Vallar and Bolognini, 2014). However, there is no known condition of full multi-sensory impairment in perception, due to the sheer rarity that all primary sensory cortices (or key multi-sensory integration regions) should be affected in unison by neurological damage or disease. The existence of multi-sensory aphantasia therefore marks a crucial difference between perception and imagery, emerging from the distinct dynamics of each process – in imagery, but not in perception, it is possible for all conscious multi-sensory experience to be abolished.

Another possible reason for this global absence of sensory mental imagery could be that the top-down signals in multi-sensory aphantasia do indeed elicit representations in various downstream sensory regions, but that some higher-order metacognitive mechanism is deficient to the extent that these subjects cannot introspect on this activity. This would be a case of intact but "unconscious" imagery – similar to the case of blindsight in perception. It is a hypothesis that is partially refuted by the absence of physiological signatures of visual imagery processes in individuals with aphantasia that would otherwise occur in subjects who report having visual imagery (such as pupillometric and skin conductance responses to attempted imagery; Wicken et al., 2021; Kay et al., 2022). However, more recent work (which has not yet undergone peer review) found that when a decoder was trained on activity in early visual cortex during a working memory delay period, the orientation of memorised gratings could be decoded in subjects with both low and high self-reported visual imagery ability (Weber et al., 2023). While this decoding accuracy correlated with visual working memory performance (irrespective of self-rated imagery vividness), it did not correlate with

imagery vividness (measured on the VVIQ) itself (although it should be said that imagery vividness is not necessarily synonymous with self-ratings of working memory performance). Even more intriguingly, another recent study (which also has not yet undergone peer review) appears to provide evidence for an intact univariate BOLD response in early visual cortex during attempted visual imagery by individuals with aphantasia, suggesting that despite self-reports of visual imagery absence, some imagery information could indeed be preserved in early visual cortex in aphantasia (Meng et al., 2023).

It is important to note, however, that the significant decoding results in both studies are based on decoders trained on activity during the imagery or working memory delay period and tested on the same delay period in left-out trials. The Weber et al. (2023) study appears not to have tested cross-decoding performance (training on perception and testing on imagery or working memory), and when this was attempted in the Meng et al. (2023) study, above-chance cross-decoding performance was only achieved in control subjects, and not in subjects with aphantasia. This suggests that although modulation of visual cortical activity may occur during attempted imagery in aphantasia, this activity may be transformed or orthogonalized such that it diverges from content-equivalent perceptual signals. This would align well with research suggesting that subjective imagery vividness is correlated with the extent of overlap between imagery and perceptual signals in the early visual stream (Dijkstra et al., 2017).

In the case of 'unconscious' imagery, multi-sensory aphantasia might also occur due to generalized differences in higher-order brain regions responsible either for generating initial multi-sensory imagery signals or for conscious readout of feedback signals from primary sensory regions. It is also possible that multi-sensory aphantasia could occur due to altered dynamics in multi-sensory integration regions of the brain, somewhere above primary sensory regions, but below prefrontal executive control regions in the mental imagery "reverse hierarchy". Very little is known about how sensory information in distributed unimodal regions is recombined to form conscious, multi-sensory imagery. However, a select few neuroimaging studies have produced evidence for "supramodal" imagery networks. These are domain-general networks that appear to be active during mental imagery, irrespective of sensory modality. These networks overlap most heavily with the default mode network (DMN), but in some cases, they also share similarities with networks involved in semantic processing (Zvyagintsev et al., 2013) or attention (such as the "task-positive" dorsal attention network; Murphy et al., 2019). From these supramodal imagery network studies, it is challenging to disentangle precisely which brain regions are involved selectively in multisensory imagery and which simply reflect non-specific, supportive cognitive mechanisms (such as attention). However, these studies, in part, indicate that conscious experience of multi-sensory imagery may be supported by a combination of domain-general and domain-specific brain networks working in tandem (or as some have suggested, a single distributed information processing network that enables both the maintenance and manipulation of sensory imagery through a task-regulated switching of connectivity profiles; Schlegel et al., 2013). Alterations or impairment in such a widespread network (or series of networks) is another candidate explanation for how the case of multi-sensory aphantasia might emerge.

Considering these theories, it is the case of visual aphantasia (rather than the multi-sensory aphantasia case) that becomes most curious. If multisensory imagery is indeed widespread in naturalistic cognition, then its deficit should be the norm rather than the exception in aphantasia, and unimodal sensory imagery deficits should be quite rare (counter to perception, where they are common). Explaining visual aphantasia in terms of a deficient top-down fronto-parietal signal or generalized metacognitive mechanism seems complex. We would either need to explain why the deficiency in universal top-down imagery signal results in selective visual imagery impairments, or we would need to invoke exclusive metacognitive feedback mechanisms for each independent sensory imagery modality (of which only the visual feedback mechanism should be impaired in visual aphantasia). Typically, sensory processing deficits in perception are the result of damage occurring early in the perceptual processing hierarchy (such as to primary sensory regions or to sensory organs themselves), impairing downstream inference. In contrast, if we assume that mental imagery works like a form of weak top-down perception (or "vision in reverse" in the case of visual imagery), where information flows downstream from higher-order regions to primary sensory regions (Dentico et al., 2014; Dijkstra et al., 2017; Dijkstra et al., 2020), visual aphantasia would have to result from changes in the downstream region itself (rather than selectively impaired visual metacognition, or solely deficient top-down signals from higher-order regions).

There is indeed some evidence for such a 'natural' explanation of visual aphantasia. In subjects with visual imagery, V1 surface size, resting cortical excitability in V1, and the ratio of V1 to superior frontal activity all appear to be associated with sensory imagery strength and precision (Bergmann et al., 2016; Keogh et al., 2020). Individuals with aphantasia also score at floor on behavioural measures of visual imagery strength thought to rely on the interaction between imagery and perception in early visual regions (Keogh and Pearson, 2018). Compared to individuals with moderate to strong imagery, individuals with aphantasia appear to show reduced resting state connectivity between prefrontal regions and visual-occipital networks (Milton et al., 2021), a finding bolstered in part by more recent unpublished neuroimaging research (Liu et al., 2023). It is possible, then, that visual aphantasia is a special case of top-down "mental blindness" characterised by altered primary visual cortical activity, specific altered connectivity between visual areas and higher-order association or executive control regions, or a combination of these two factors. In these individuals with selective visual aphantasia, preserved top-down connections to non-visual sensory regions would still enable mental imagery in all non-visual sensory modalities, matching the self-reports of individuals in our study with visual aphantasia but preserved imagery vividness in other modalities.

Detailed future research will be needed to uncover the precise mechanisms underpinning multi-sensory aphantasia and visual aphantasia, respectively. It will also be important to clarify whether there are empirical differences in cognitive functioning between these two subpopulations. Answering these questions will ultimately require both carefully designed neuroimaging paradigms which clarify the networks involved in multi-sensory imagery, as well as behavioural paradigms which can dissociate mental imagery performance from higher-order metacognitive imagery processes. In light of this, several limitations of our current dataset should be acknowledged. Most importantly, our data may be affected by a self-selection bias, since individuals who participated in our studies volunteered on the basis of self-identified visual aphantasia (given that aphantasia is primarily known as a visual phenomenon in the public media). A similar issue affects all previous studies screening for aphantasia using the VVIQ alone – although this is standard practice in the aphantasia field, it is now clearer than ever that this metric may disproportionately sample subjects who primarily report a lack of visual imagery (rather than an absence of imagery in another domain, such as auditory imagery). Hinwar and Lambert (2021) have established the co-occurrence of visual aphantasia amongst individuals screened based on reported auditory (rather than visual) imagery deficits (and vice versa). This is important as it demonstrates that multi-sensory imagery deficits are often reported irrespective of the sampling criterion. However, the potential sampling bias that media coverage on pure visual aphantasia might have caused for the aphantasia field remains to be seen.

Solutions to these problems might include screening aphantasia based on multi-sensory imagery questionnaires instead (such as the QMI), and more formally estimating the co-occurrence of visual aphantasia in datasets screened using non-visual imagery questionnaires, such as the sensory modality-specific imagery questionnaires used by Hinwar and Lambert (2021) and Dance et al. (2021). In future, it may also be necessary to begin using a combined questionnaire and

behavioural task cut-off (as suggested by Wicken et al., 2021) to ensure that aphantasic subjects who report weak imagery vividness also perform at floor on behavioural imagery measures. This will help to differentiate between cases where aphantasia has a primarily meta-cognitive origin (measured by low self-reported imagery vividness but intact task performance) and cases where aphantasia is indeed marked by sensory imagery absence due to alterations to fronto-parietal regions, early sensory processing regions, or connectivity between the two.

Lastly, we cannot draw any inferences from our questionnaire-based dataset as to how different aphantasia sub-groups might differ on cognitive tasks involving imagery. It will be important for future research to explore whether individuals with selective visual aphantasia perform better on average than those with multi-sensory aphantasia on cognitive tasks requiring multi-sensory imagery strategies. Current behavioural measures of visual imagery strength (such as the binocular rivalry task: Pearson et al., 2008; Keogh and Pearson, 2018; or associated physiological indices: Wicken et al., 2021; Kay et al., 2022) cannot differentiate individuals with multi-sensory aphantasia from those with visual aphantasia given the visual nature of the tasks. It is therefore important that novel multi-sensory and cross-modal imagery tasks are designed which better measure the inherently multi-sensory nature of mental imagery. This will enable us to build models of multi-sensory integration during mental imagery, draw a clearer taxonomy of aphantasia and its diversity, and better understand crucial differences in the neural architecture supporting sensory perception and mental simulation.

Overall, our research reveals that aphantasia is a heterogeneous phenomenon characterised by distinct sub-groups of individuals with various patterns of self-reported multi-sensory imagery impairments. Whilst pure visual aphantasia appears to comprise a large portion (30%) of this sub-population, at least 24% of individuals with aphantasia report total multi-sensory imagery absence. Unique patterns of multi-sensory imagery reports also exist, with some individuals only reporting mental imagery loss in two selective sensory modalities, and with other individuals only reporting preserved mental imagery in a single sensory modality. These findings suggest that aphantasia may be underpinned by diverse aetiologies, each of which may lead to vastly different consequences for our understanding of network-level mental imagery processes in the human brain. These factors should be carefully accounted for in future imagery research, and there is a general onus on researchers to consider how multi-sensory integration might occur during naturalistic, everyday mental imagery. A growing body of research has demonstrated the breadth of inter-individual variation in visual imagery capacity in the general population, positioning aphantasia in a unique and informative location on this spectrum and emphasising group differences between those with imagery and those without. Our current findings emphasise that there is also a large diversity of cognitive reports within the aphantasia sub-population itself, whose subjective experiences may teach us invaluable lessons about how the human brain allows us to perceive and imagine the world around us.

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Author contributions

All authors developed the study concept and design. AJD and RK collected the data, and AJD performed data analysis. AJD drafted the first manuscript, and RK and JP provided critical revisions. All the authors approved the final manuscript for submission.

Declaration of Interests

The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.neures.2023.11.009.

References

Albers, A.M., Kok, P., Toni, I., Dijkerman, H.C., de Lange, F.P., 2013. Shared representations for working memory and mental imagery in early visual cortex. Curr. Biol. 23 (15), 1427–1431. https://doi.org/10.1016/j.cub.2013.05.065.

Andrade, J., May, J., Deeprose, C., Baugh, S.-J., Ganis, G., 2014. Assessing vividness of mental imagery: the plymouth sensory imagery questionnaire. Br. J. Psychol. 105 (4), 547–563. https://doi.org/10.1111/bjop.12050.

Avery, J.A., Carrington, M., Martin, A., 2023. A common neural code for representing imagined and inferred tastes. Prog. Neurobiol. 223, 102423 https://doi.org/10.1016/j.pneurobio.2023.102423.

Berger, C.C., Ehrsson, H.H., 2013. Mental imagery changes multisensory perception. Curr. Biol. 23 (14), 1367–1372. https://doi.org/10.1016/j.cub.2013.06.012.

Bergmann, J., Genç, E., Kohler, A., Singer, W., Pearson, J., 2016. Smaller primary visual cortex is associated with stronger, but less precise mental imagery. Cereb. Cortex 26 (9), 3838–3850. https://doi.org/10.1093/cercor/bhv186.

Calvert, G.A., Spence, C., & Stein, B.E.. (Eds.). , 2004. The handbook of multisensory processes (pp. xvii, 915). Boston Review. https://doi.org/10.7551/mitpress/3422.001.0001.

Cui, X., Jeter, C.B., Yang, D., Montague, P.R., Eagleman, D.M., 2007. Vividness of mental imagery: individual variability can be measured objectively. Vis. Res. 47 (4), 474–478. https://doi.org/10.1016/j.visres.2006.11.013.

Dance, C.J., Ward, J., Simner, J., 2021. What is the link between mental imagery and sensory sensitivity? insights from aphantasia. Perception 50 (9), 757–782. https://doi.org/10.1177/03010066211042186.

Dawes, A.J., Keogh, R., Andrillon, T., Pearson, J., 2020. A cognitive profile of multi-sensory imagery, memory and dreaming in aphantasia. Nat. *Sci. Rep.* 10 https://doi.org/10.1038/s41598-020-65705-7.

Dawes, A.J., Keogh, R., Robuck, S., Pearson, J., 2022. Memories with a blind mind: remembering the past and imagining the future with aphantasia. Cognition 227, 105192. https://doi.org/10.1016/j.cognition.2022.105192.

de Borst, A.W., de Gelder, B., 2017. fMRI-based multivariate pattern analyses reveal imagery modality and imagery content specific representations in primary somatosensory, motor and auditory cortices. Cereb. Cortex 27 (8), 3994–4009. https://doi.org/10.1093/cercor/bhw211.

Dentico, D., Cheung, B.L., Chang, J.-Y., Guokas, J., Boly, M., Tononi, G., Van Veen, B., 2014. Reversal of cortical information flow during visual imagery as compared to visual perception. NeuroImage 100, 237–243. https://doi.org/10.1016/j.neuroimage.2014.05.081.

Dijkstra, N., Bosch, S., van Gerven, M.A.J., 2017. Vividness of visual imagery depends on the neural overlap with perception in visual areas. J. Neurosci. https://doi.org/10.1523/JNEUROSCI.3022-16.2016.

Dijkstra, N., Ambrogioni, L., Vidaurre, D., van Gerven, M., 2020. Neural dynamics of perceptual inference and its reversal during imagery. eLife 9, e53588. https://doi.org/10.7554/eLife.53588.

Dijkstra, N., Zeidman, P., Ondobaka, S., van Gerven, M. a J., Friston, K., 2017. Distinct top-down and bottom-up brain connectivity during visual perception and imagery. Sci. Rep. 7 (1), 5677 https://doi.org/10.1038/s41598-017-05888-8.

Ehrsson, H.H., Geyer, S., Naito, E., 2003. Imagery of voluntary movement of fingers, toes, and tongue activates corresponding body-part-specific motor representations. J. Neurophysiol. 90 (5), 3304–3316. https://doi.org/10.1152/jn.01113.2002.

Hinwar, R.P., Lambert, A.J., 2021. Anauralia: the silent mind and its association with aphantasia. Front. Psychol. 12. ⟨https://www.frontiersin.org/article/10.3389/fpsyg.2021.744213⟩.

Huang, Y., Pollick, F., Liu, M., Zhang, D., 2023. Gabor and non-gabor neural representations are shared between visual perception and mental imagery. J. Cogn. Neurosci. 1–16. https://doi.org/10.1162/jocn_a_01992.

Hubbard, T.L., 2010. Auditory imagery: empirical findings. Psychol. Bull. 136 (2), 302–329. https://doi.org/10.1037/a0018436.

Ishai, A., Haxby, J.V., Ungerleider, L.G., 2002. Visual imagery of famous faces: effects of memory and attention revealed by fMRI. NeuroImage 17 (4), 1729–1741. https://doi.org/10.1006/nimg.2002.1330.

Johnson, M., Johnson, M., 2014. Decoding individual natural scene representations during perception and imagery. Front. Hum. Neurosci. 8, 59. https://doi.org/10.3389/fnhum.2014.00059.

Kamitani, Y., Tong, F., 2006. Decoding seen and attended motion directions from activity in the human visual cortex (CB). Curr. Biol. 16 (11), 1096–1102. https://doi.org/10.1016/j.cub.2006.04.003.

Kasai, T., Kawai, S., Kawanishi, M., Yahagi, S., 1997. Evidence for facilitation of motor evoked potentials (MEPs) induced by motor imagery. Brain Res. 744 (1), 147–150. https://doi.org/10.1016/s0006-8993(96)01101-8.

Kay, L., Keogh, R., Andrillon, T., Pearson, J., 2022. The pupillary light response as a physiological index of aphantasia, sensory and phenomenological imagery strength. e72484 eLife 11. https://doi.org/10.7554/eLife.72484.

Keogh, R., Pearson, J., 2018. The blind mind: no sensory visual imagery in aphantasia. Cortex 105, 53–60 https://doi.org/10.1016/J.CORTEX.2017.10.012.

Keogh, R., Bergmann, J., Pearson, J., 2020. Cortical excitability controls the strength of mental imagery. e50232 eLife 9. https://doi.org/10.7554/eLife.50232.

Keogh, R., Wicken, M., Pearson, J., 2021. Visual working memory in aphantasia: retained accuracy and capacity with a different strategy. Cortex 143, 237–253. https://doi.org/10.1016/j.cortex.2021.07.012.

Koenig-Robert, R., Pearson, J., 2019. Decoding the contents and strength of imagery before volitional engagement. Sci. Rep. 9 (1), 3504 https://doi.org/10.1038/s41598-019-39813-y.

Koenig-Robert, R., Pearson, J., 2021. Why do imagery and perception look and feel so different? Philos. Trans. R. Soc. B: Biol. Sci. 376 (1817), 20190703. https://doi.org/10.1098/rstb.2019.0703.

Liu, J., Zhan, M., Hajhajate, D., Spagna, A., Dehaene, S., Cohen, L., Bartolomeo, P. , 2023. Ultra-high field fMRI of visual mental imagery in typical imagers and aphantasic individuals (p. 2023.06.14.544909). bioRxiv. https://doi.org/10.1101/2023.06.14.544909.

Meng, M., Chang, S., Zhang, X., Pearson, J., 2023. Imageless imagery in aphantasia: decoding non-sensory imagery in aphantasia. Unpubl. (Peer Rev.). https://doi.org/10.21203/rs.3.rs-3162223/v1.

Milton, F., Fulford, J., Dance, C., Gaddum, J., Heuerman-Williamson, B., Jones, K., Knight, K.F., MacKisack, M., Winlove, C., Zeman, A., 2021. Behavioral and neural signatures of visual imagery vividness extremes: aphantasia versus hyperphantasia. Cereb. Cortex Commun. 2 (2) https://doi.org/10.1093/texcom/tgab035.

Murphy, C., Rueschemeyer, S.-A., Smallwood, J., Jefferies, E., 2019. Imagining sounds and images: Decoding the contribution of unimodal and transmodal brain regions to semantic retrieval in the absence of meaningful input. J. Cogn. Neurosci. 31 (11), 1599–1616. https://doi.org/10.1162/jocn_a_01330.

Naselaris, T., Olman, C.A., Stansbury, D.E., Ugurbil, K., Gallant, J.L., 2015. A voxel-wise encoding model for early visual areas decodes mental images of remembered scenes. NeuroImage 105, 215–228. https://doi.org/10.1016/j.neuroimage.2014.10.018.

Nierhaus, T., Wesolek, S., Pach, D., Witt, C.M., Blankenburg, F., Schmidt, T.T., 2023. Content representation of tactile mental imagery in primary somatosensory cortex. eNeuro 10 (6). https://doi.org/10.1523/ENEURO.0408-22.2023.

O'Craven, K.M., Kanwisher, N., 2000. Mental imagery of faces and places activates corresponding stiimulus-specific brain regions. J. Cogn. Neurosci. 12 (6), 1013–1023. https://doi.org/10.1162/08989290051137549.

Pearson, J., Kosslyn, S.M., 2015. The heterogeneity of mental representation: ending the imagery debate. Proc. Natl. Acad. Sci. 112 (33), 10089–10092. https://doi.org/10.1073/pnas.1504933112.

Pearson, J., Clifford, C.W.G., Tong, F., 2008. The functional impact of mental imagery on conscious perception. Curr. Biol. 18 (13), 982–986. https://doi.org/10.1016/j.cub.2008.05.048.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., 2011. Scikit-learn: machine learning in python. J. Mach. Learn. Res. 12, 2825–2830.

Pounder, Z., Jacob, J., Evans, S., Loveday, C., Eardley, A.F., Silvanto, J., 2022. Only minimal differences between individuals with congenital aphantasia and those with typical imagery on neuropsychological tasks that involve imagery. Cortex 148, 180–192. https://doi.org/10.1016/j.cortex.2021.12.010.

Proix, T., Delgado Saa, J., Christen, A., Martin, S., Pasley, B.N., Knight, R.T., Tian, X., Poeppel, D., Doyle, W.K., Devinsky, O., Arnal, L.H., Mégevand, P., Giraud, A.-L., 2022. Imagined speech can be decoded from low- and cross-frequency intracranial EEG features. *Nat. Commun.*, *13*(1), Artic. 1. https://doi.org/10.1038/s41467-021-27725-3.

Schlegel, A., Kohler, P.J., Fogelson, S.V., Alexander, P., Konuthula, D., Tse, P.U., 2013. Network structure and dynamics of the mental workspace. Proc. Natl. Acad. Sci. 110 (40), 16277–16282. https://doi.org/10.1073/pnas.1311149110.

Sheehan, P.W., 1967. A shortened form of Betts' questionnaire upon mental imagery. J. Clin. Psychol. 23 (3), 386–389. https://doi.org/10.1002/1097-4679(196707)23:3<386::AID-JCLP2270230328>3.0.CO;2-S.

Slotnick, S.D., Thompson, W.L., Kosslyn, S.M., 2005. Visual mental imagery induces retinotopically organized activation of early visual areas. Cereb. Cortex (N. Y., N. Y.: 1991) 15 (10), 1570–1583. https://doi.org/10.1093/cercor/bhi035.

Stokes, M., Thompson, R., Cusack, R., Duncan, J., 2009. Top-down activation of shape-specific population codes in visual cortex during mental imagery. J. Neurosci.: Off. J. Soc. Neurosci. 29 (5), 1565–1572. https://doi.org/10.1523/JNEUROSCI.4657-08.2009.

Takahashi, J., Saito, G., Omura, K., Yasunaga, D., Sugimura, S., Sakamoto, S., Horikawa, T., Gyoba, J., 2023. Diversity of aphantasia revealed by multiple assessments of visual imagery, multisensory imagery, and cognitive style. Front. Psychol. 14, 1174873 https://doi.org/10.3389/fpsyg.2023.1174873.

Vallar, G., Bolognini, N. , 2014. Unilateral Spatial Neglect. In A. C. (Kia) Nobre & S. Kastner (Eds.), The Oxford Handbook of Attention (p. 0). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199675111.013.012.

van der Maaten, L., Hinton, G., 2008. Visualizing data using t-SNE. J. Mach. Learn. Res. 1 (1–48). ⟨https://www.jmlr.org/papers/volume9/vandermaaten08a/vandermaaten08a.pdf⟩.

Vetter, P., Smith, F.W., Muckli, L., 2014. Decoding sound and imagery content in early visual cortex. Curr. Biol. 24 (11), 1256–1262. https://doi.org/10.1016/j.cub.2014.04.020.

Weber, S., Christophel, T., Görgen, K., Soch, J., Haynes, J.-D.2023. Working memory signals in early visual cortex do not depend on visual imagery (p. 2023.02.13.528298). bioRxiv (Unpublished, Under Peer Review). https://doi.org/10.1101/2023.02.13.528298.

Wicken, M., Keogh, R., Pearson, J., 2021. The critical role of mental imagery in human emotion: insights from fear-based imagery and aphantasia, 20210267–20210267 Proc. R. Soc. B: Biol. Sci. 288 (1946). https://doi.org/10.1098/rspb.2021.0267.

Yoo, S.-S., Freeman, D.K., McCarthy, J.J.I., Jolesz, F.A., 2003. Neural substrates of tactile imagery: a functional MRI study. NeuroReport 14 (4), 581.

Zatorre, R.J., Halpern, A.R., 2005. Mental concerts: musical imagery and auditory cortex. Neuron 47 (1), 9–12. https://doi.org/10.1016/j.neuron.2005.06.013.

Zeman, A., Dewar, M., Della Sala, S., 2015. Lives without imagery—congenital aphantasia. a J. Devoted Study Nerv. Syst. Behav. 73, 378–380. https://doi.org/10.1016/j.cortex.2015.05.019.

Zeman, A., Milton, F., Della Sala, S., Dewar, M., Frayling, T., Gaddum, J., Hattersley, A., Heuerman-Williamson, B., Jones, K., MacKisack, M., Winlove, C., 2020. Phantasia–the psychological significance of lifelong visual imagery vividness extremes. Cortex 130, 426–440. https://doi.org/10.1016/j.cortex.2020.04.003.

Zeman, A.Z.J., Della Sala, S., Torrens, L.A., Gountouna, V.-E., McGonigle, D.J., Logie, R. H., 2010. Loss of imagery phenomenology with intact visuo-spatial task performance: a case of 'blind imagination. Neuropsychologia 48 (1), 145–155. https://doi.org/10.1016/j.neuropsychologia.2009.08.024.

Zhao, B., Della Sala, S., Zeman, A., Gherri, E., 2022. Spatial transformation in mental rotation tasks in aphantasia. Psychon. Bull. Rev. https://doi.org/10.3758/s13423-022-02126-9.

Zvyagintsev, M., Clemens, B., Chechko, N., Mathiak, K.A., Sack, A.T., Mathiak, K., 2013. Brain networks underlying mental imagery of auditory and visual information. Eur. J. Neurosci. 37 (9), 1421–1434. https://doi.org/10.1111/ejn.12140.